

Linux Kernel Learning Roadmap

Next steps after building your own custom filesystem

You've written a filesystem, so you already know the VFS contract from the outside. This roadmap takes you inward — from how the kernel caches and writes your data, down through the block layer, the concurrency primitives that keep it correct, and the journaling that keeps it crash-safe. Read real filesystems alongside each topic.

1. VFS Layer (deepen)

- `struct super_block`, `inode`, `dentry`, `file` and their operation tables (`super_operations`, `inode_operations`, `file_operations`, `dentry_operations`)
- Dentry cache (dcache) and inode cache mechanics
- Mount namespaces, `vfsmount`, bind mounts

2. Page Cache & Memory

- `address_space` / `address_space_operations` (`readpage`, `writepage`, `write_begin` / `write_end`)
- Page cache, dirty page writeback, writeback threads
- mmap path: `vm_operations_struct`, fault handling
- Folios — the modern replacement for individual pages

3. Block Layer & I/O

- `bio` structure, request queues, I/O schedulers (mq-deadline, BFQ)
- `blk-mq` multi-queue architecture
- buffer heads vs. bio-based I/O

4. Journaling & Consistency

- JBD2 — study how ext4 uses it
- Write barriers, FUA/flush, crash consistency
- `fsync` semantics

5. Concurrency

- Locking: spinlocks, mutexes, RW semaphores, seqlocks
- RCU (read-copy-update) — heavily used in VFS/dcache
- Per-CPU data, memory barriers

6. Study Real Filesystems (read source)

- `ramfs` / `tmpfs` — simplest, in-memory
- `ext4` — journaling, extents, real-world complexity
- `f2fs` — log-structured, flash-oriented
- `overlayfs` — stacking
- FUSE internals if you want a userspace comparison

7. Tooling & Debugging

- `ftrace`, `perf`, `bpftool` / eBPF for tracing FS paths
- KASAN, lockdep, kmemleak
- `xfstests` for filesystem testing
- `blktrace` for block I/O analysis

8. Recommended Reading

- *Linux Kernel Development* (Love) — foundation
- *Understanding the Linux Kernel* (Bovet & Cesati) — VFS/memory depth
- Kernel docs: [Documentation/filesystems/vfs.rst](#)
- LWN.net articles on filesystems and folios

Suggested Order

VFS internals → **page cache** → **block layer** → **concurrency (RCU/locking)** → **journaling**

Read `ramfs` first, then `ext4` alongside the later topics.